

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

Methods of data collection

Data collection methods are techniques used to gather information for research, decision-making, or analysis, ranging from surveys and interviews to observation and experimentation. Key approaches include primary data collection and secondary data, and analysis (using existing information). These methods can be either quantitative (numerical) or qualitative (descriptive)

- **Data:** Data is a tool that helps an investigator in understanding the problem by providing him with the information required.
- **Data Collection** is the systematic process of gathering, measuring and analyzing information from various sources to gain an accurate understanding of a specific topic or problem.
- **Survey:** A survey is a method of collecting information from a group of individuals to study characteristics such as quality, price, usefulness, satisfaction, etc.

The main objectives of data collection are:

- To support decision-making.
- To identify trends and patterns.
- To measure performance and progress.
- To provide evidence for conclusions

Primary Data

Primary data is information collected directly from original sources for a specific research purpose. It is fresh, relevant and tailored to the study.

Advantages of primary data is

- 1.High accuracy
- 2.More control over data quality
- 3.Specific to research objectives

■ Some common methods for collecting primary data:

1. Interviews: Interviews involve direct communication between the investigator and respondents.

Advantage: Provides real-time, natural data; no reliance on self-reported information.

Disadvantage: Observer bias; limited to what can be seen; may influence subjects' behavior

2. Questionnaires: is a structured set of questions prepared to collect information. The investigator can collect data through the questionnaire in two ways:

Mailing Method: Questionnaires are sent by mail or online.

Enumerator's Method: The enumerator personally visits respondents and fills the questionnaire.

Questionnaire can takes several forms such as

A) Closed-Ended Questionnaire.

B) Open-Ended Questionnaire

Advantage: Can reach a large audience quickly and cost-effectively.

Disadvantage: Responses may be biased or inaccurate; low response rates.

3. Observations: The method involves collecting data by watching and recording behaviors or events as they naturally occur.

Advantage: Provides real-time, authentic data without reliance on self-reported information.

Disadvantage: Risk of observer bias and behavior changes.

4. Experiments: This method involves manipulating variables in a controlled environment to study cause-and-effect relationships.

Advantage: Allows for the establishment of cause-and-effect relationships with high precision.

Disadvantage: Can be expensive and less realistic.

5. Focus Group: A focus group gathers 6–12 participants to discuss a topic under a moderator's guidance.

Advantage: Provides diverse and detailed insights.

Disadvantage: Results may not represent the larger population

6. Local Correspondents: In Local Correspondent method, for the collection of data, the investigator appoints correspondents or local persons at various places, which are then furnished by them to the investigator. With the help of correspondents and local persons, the investigators can cover a wide area.

■ Secondary Data

Secondary data is collected from information that has already been gathered, processed and published by others. It is broadly classified into Published Sources and Unpublished Sources.

Methods of Collecting Secondary Data: can be collected through different published and unpublished sources. Some of them are as follows:

1. Published Sources: are officially available reports and documents that provide reliable and structured data for research and analysis.

Government Publications: Central and State Governments publish statistical reports such as census data.

Semi-Government Publications: Different Semi-Government bodies also publish data related to health, education, deaths and births. These kinds of data are also reliable and used by different informants.

- **Journals and Papers:** Different newspapers and magazines provide a variety of statistical data in their writings, which are used by different investigators for their studies.
- **International Publications:** Different international organizations like World Bank, etc.

- **Publications of Research Institutions:** Research institutions and universities also publish their research activities and their findings, which are used by different investigators as secondary data.

2. Unpublished Sources

Unpublished sources are another source of collecting secondary data. The data in unpublished sources is collected by different government organizations and other organizations. These organizations usually collect data for their self-use and are not published anywhere. For example, research work done by professors, professionals, teachers and records maintained by private enterprises.

Sources of Data:

1) Routinely kept records: formal, consistent documents created and maintained to track the history of a person's or organization's activities, often used for operational, legal, or financial purposes. These records are essential for ensuring continuity in care, maintaining compliance with regulations, and acting as evidence in potential disputes

For example: Hospital medical records contain immense amounts of information on patients.

2) External sources.

The data needed to answer a question may already exist in the form of Published reports, commercially available data banks, or the research literature,

i.e. someone else has already asked the same question

3) Surveys:

The source may be a survey, if the data needed is about answering certain questions.

For example: If the administrator of a clinic wishes to obtain information regarding the mode of transportation used by patients to visit the clinic, then a survey may be conducted among patients to obtain this information

4) Experiments:

Frequently the data needed to answer a question are available only as the result of an experiment.

For example: If a nurse wishes to know which of several strategies is best for maximizing patient compliance, she might conduct an experiment in which the different strategies of motivating compliance are tried with different patients.

Collecting data technique:

1. The first thing need to do is choose what details wanted to be collect.
2. Decide what topics the information will cover.
3. Who want to collect it from and how much data will be needed.
- 4.The goals needed to hope when use data
5. Determine the answers to these questions

Thank

You

